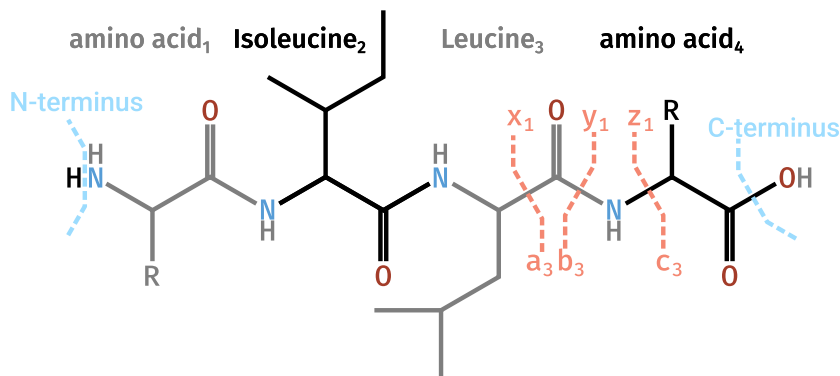


Backbone fragmentation^{1,2}



	Formula ($h = H^+ \times z$)	Mass ($h = 1.008 \times z$)
a	$\sum(AA) - CO + h$	$\sum(AA) - 27.995 + h$
b	$\sum(AA) + h$	$\sum(AA) + h$
c	$\sum(AA) + NH_3 + h$	$\sum(AA) + 17.027 + h$
d	$\sum(AA) + C_2H_5N + h$	$\sum(AA) + 41.027 + h$
v	$\sum(AA) + C_2H_5NO + h$	$\sum(AA) + 56.014 + h$
w	$\sum(AA) + C_3H_5O + h$	$\sum(AA) + 55.018 + h$
x	$\sum(AA) + CO - H_2 + h$	$\sum(AA) + 25.979 + h$
y	$\sum(AA) + H_2O + h$	$\sum(AA) + 18.011 + h$
z	$\sum(AA) + O - NH + h$	$\sum(AA) + 0.984 + h$

1,2,3

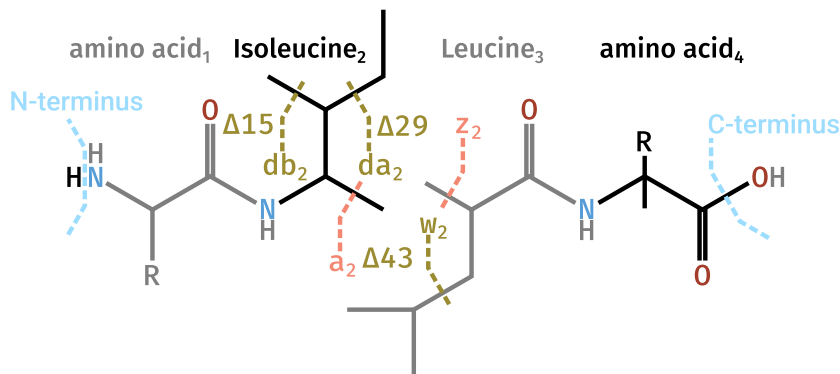
Variant ions

	"/-2	"/-1	Base	"/+1	"/+2
a			aAU	U	U
b			aAEhU		U
c		e	aAeEU	E	e
d			aAU		
v			aAeU		
w		e	aAeEU	e	
x		U	aAU	U	U
y	U	U	aAeEU		
z		e	aAeEU	eE	e

Variant ions are ion series differing by one or two hydrogen atoms in mass from the main ion series. The occurrence of these is dependent on the fragmentation technique used.

a=EAD
A=EACID
e=ExD⁴⁻⁸
E=EtHCD
h=HCD
U=UVPD⁹⁻¹²

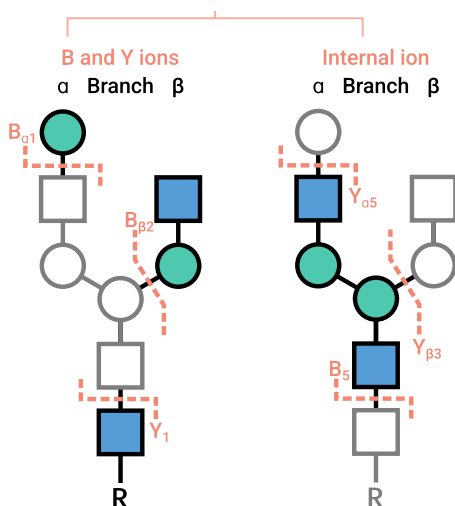
Side chain fragmentation / satellite ions^{2,3}



Satellite ions are formed as secondary fragmentation, d from an a ion, w from a z ion. For Threonine and Isoleucine there are two possible d and w ions. These can be listed as 'd/wa_n' and 'd/wb_n' where a is the heaviest of the two options. The leaving groups for Threonine are CH₃ for wa and OH for wb, for Isoleucine the leaving groups are CH₃ for wa and C₂H₅ for wb. Glycine, alanine, and proline have no satellite ions. The v ion is full loss of the sidechain from a y ion. For ETD and ECD, side chain loss similar to v ions can be observed from many fragments possibly even with multiple side chains lost⁴. Additionally in ECD, w fragmentation can be found in sidechains away from the backbone cleavage^{5,6}. Denoted with ^dw, where d indicates the distance from the backbone cleavage, ⁰w, the standard w ion, is written as 'w'.

Glycan¹³

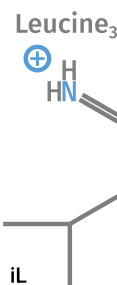
Collectively B and Internal ions are known as Oxonium ions



Ion	Formula	Mass	Ion	Formula	Mass
H-C ₂ H ₆ O ₃	C ₄ H ₅ O ₂	85.0284	S-H ₂ O	C ₁₁ H ₁₆ NO ₇	274.0921
H-CH ₆ O ₃	C ₅ H ₅ O ₂	97.0284	S	C ₁₁ H ₁₈ NO ₈	292.1027
H-2H ₂ O	C ₆ H ₅ O ₃	127.0390	G-H ₂ O	C ₁₁ H ₁₆ NO ₈	290.0870
H-H ₂ O	C ₆ H ₆ O ₄	145.0495	G	C ₁₁ H ₁₈ NO ₉	308.0976
H	C ₆ H ₁₁ O ₅	163.0601	H+N	C ₁₄ H ₂₄ NO ₁₀	366.1395
N-C ₂ H ₆ O ₄	C ₄ H ₆ NO	84.0444	2N	C ₁₆ H ₂₇ N ₂ O ₁₀	407.1660
N-C ₂ H ₆ O ₃	C ₆ H ₆ NO ₂	126.0550	H+2N	C ₂₂ H ₃₇ N ₂ O ₁₅	569.2188
N-CH ₆ O ₃	C ₇ H ₆ NO ₂	138.0550	H+N+S	C ₂₅ H ₄₁ N ₂ O ₁₈	657.2349
N-C ₂ H ₄ O ₂	C ₆ H ₁₀ NO ₃	144.0655	H=Hex		
N-2H ₂ O	C ₈ H ₁₀ NO ₃	168.0655	N=HexNAc		
N-H ₂ O	C ₈ H ₁₂ NO ₄	186.0761	S=Neu5Ac		
N	C ₈ H ₁₄ NO ₅	204.0867	G=Neu5Gc		

Immonium

Formed by double backbone cleavage. Commonly fragments further by neutral loss. These fragments indicate existence of certain amino acids, but not all amino acids form immonium ions. The immonium ions have a mass between 0 and 160 Da. Formula: amino acid - CO + H

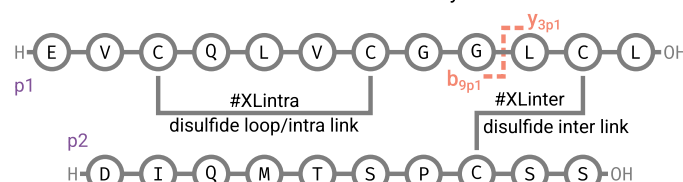


Amino Acid	Formula	Mass	Amino Acid	Formula	Mass
A	C ₂ H ₆ N	44.050	K	C ₅ H ₁₃ N ₂	101.108
R	C ₅ H ₁₃ N ₄	129.114	M	C ₄ H ₁₀ NS	104.053
N	C ₃ H ₇ N ₂ O	87.056	F	C ₆ H ₁₀ N	120.081
D	C ₃ H ₆ NO ₂	88.040	P	C ₄ H ₈ N	70.066
C	C ₂ H ₆ NS	76.022	S	C ₂ H ₆ NO	60.045
E	C ₄ H ₈ NO ₂	102.056	T	C ₃ H ₈ NO	74.061
Q	C ₄ H ₉ N ₂ O	101.071	W	C ₁₀ H ₁₁ N ₂	159.092
G	CH ₄ N	30.034	Y	C ₈ H ₁₀ NO	136.076
H	C ₅ H ₈ N ₃	110.072	V	C ₄ H ₁₀ N	72.081
I/L/J	C ₅ H ₁₂ N	86.097			

Formula & Mass are MH+

Cross-links

Note that if you want to see breaking cross-linkers you will need to define your own custom linkers in the Annotator as the breaking is not defined in any of the used modification databases.



Two possible masses for y_{3p1}

y_{3p1} - #XLinter LCL + OH - H_(broken disulfide)
y_{3p1} - #XLinter LCL + OH - 2H_(intact disulfide) + p₂

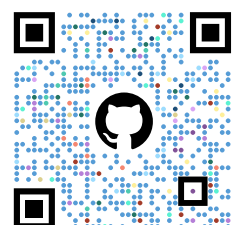
One possible mass for b_{9p1}

b_{9p1} - #XLintra EVCQLVCGG + H - 2H_(disulfide)

This disulfide is a loop link, so it could break either in the linker or in any of the contained amino acids without changing the final mass of the fragment.

References

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- [6] 10.1021/ac0619332
- [7] 10.1002/jms.3919
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- [10] 10.1007/s13361-015-1297-5
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- [12] 10.1021/acs.chemrev.9b00440
- [13] 10.1007/BF01049915
- [14] 10.1016/j.trac.2018.09.007



To annotate these and more on your own data see:
github.com/snijderlab/annotator

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V1.1

